

Equality-visibility depth in equational presentations: finite-ground spectra, certificates, and semigroup space

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Abstract

A finite equational presentation determines both an equality theory and the resources required to recover that equality on a bounded set of terms. We study the least horizon $\delta_E(d)$ at which congruence reasoning becomes exact on closed terms of depth at most d , and the pairwise threshold $\lambda_E(s, t)$. For finite ground presentations, subterm locality gives $\delta_E(d) \leq \max\{d, h(E)\}$, where $h(E)$ is the maximum axiom-side depth. The possible low-depth profiles at exact axiom depth h are precisely the weakly increasing sequences $a_0 \leq \dots \leq a_{h-1}$ with $i \leq a_i \leq h$; there are C_{h+1} such profiles. An incrementally activated query-subterm DAG computes the complete finite-query filtration. Chronological proof DAGs and finite threshold countermodels provide complementary certificates of the exact horizon. We apply this construction to the incomplete-action interface of Paper I.

We distinguish this congruence-proof resource from the maximum depth in a sequential rewrite derivation, denoted by λ_E^{rw} and δ_E^{rw} . Explicit examples separate the two, even for finite ground equations. For finite universal presentations, computability of either visibility function is equivalent to decidability of the ground word problem. Universal action-chain identities reproduce semigroup *rewrite* space exactly on typed chains. In the one-sorted language, skeleton invariance yields the exact formula $\delta_{\mathcal{A}(P)}^{\text{rw}}(n) = \max_{1 \leq m \leq n} (n - m + s_P(m))$. This separates the finite-ground certificate theory from the sequential space results while retaining their common equational semantics.

Keywords. Equational logic; congruence closure; equality visibility; ground presentations; proof certificates; term rewriting; semigroup derivational space.

1 Introduction

Paper I [1] asks what an incomplete action specification forces: which carrier elements become equal, which missing values are already determined, and which quotients admit carrier-valued completions. The present paper asks when those consequences become visible if the syntax available to a proof is bounded. This is a question about the chosen presentation, not just its final equality relation.

A small example captures the distinction. Let

$$E = \{p = f^2(r), q = f^2(r)\}.$$

Then $p = q$ follows through a term of depth two. At horizon zero or one, neither input equation is available. Adding the derived equation $p = q$ changes no full equational consequence, but makes that equality available already at horizon zero. The full theory is the same; its visibility filtration is different.

There is a second distinction that must be fixed before measuring a proof. One can bound the terms occurring in a proof DAG that uses reflexivity, symmetry, transitivity, and congruence.

Alternatively, one can bound the whole terms in a sequential derivation, where each step replaces one instance of an input equation inside a context. These proof formats give the same unrestricted equality, but their bounded versions differ. For the example above, a congruence proof of $f(p) = f(q)$ first proves $p = q$ at depth two and then applies f to the endpoints. Its maximum term depth is two. A sequential derivation must pass through $f^3(r)$ and has minimum depth three. Section 7.1 makes this separation exact.

We use λ_E and δ_E for the congruence-proof filtration, which is the resource computed by the finite-ground algorithm and used in Paper I. A superscript rw always denotes sequential rewrite visibility. The distinction is also necessary in the universal regime: semigroup derivational space measures intermediate words in a rewrite sequence, and transfers to the latter invariant.

The paper has four main contributions.

1. **A complete finite-ground spectrum.** Classical ground locality bounds $\delta_E(d)$ by $\max\{d, h(E)\}$. Together with monotonicity, it leaves a finite low-depth profile. Theorem 3.3 shows that all admissible profiles occur and that their number is Catalan. A finite extension by derived equations can eliminate the overhead altogether; the size of that extension remains a separate issue.
2. **Exact finite-query certificates.** Theorem 4.1 reduces every threshold query to one fixed input/query DAG. Incremental activation gives the exact first-equality levels. Chronological explanation capture prevents circular proofs, while finite threshold models certify that an equality was unavailable below its claimed optimum. Neither chronology alone nor a positive proof alone establishes minimality.
3. **A structural application from Paper I.** On its named action interface, the first two quotients are the sets X_1, X_2 described in Paper I, Theorem 6.6. We give the proof and lower separator for the common four-element example, whose class counts are $N_1 = 5$ and $N_2 = 3$. This also distinguishes visibility depth from the number of quotient feedback rounds.
4. **The universal and sequential boundaries.** For finite universal presentations, a computable majorant for either visibility function is equivalent to a decidable ground word problem. For sequential action-chain derivations, the typed invariant is exactly semigroup space. Theorem 8.5 shows that arbitrary one-sorted contexts replace it by the least majorant with nondecreasing overhead. The semigroup realization results are stated for δ^{rw} .

Ground congruence closure, incremental explanations, and semigroup space are established tools [3, 4, 6, 12]. Our contribution is the exact filtration and spectrum, its two-sided certificate interpretation, and the precise contextual envelope for sequential action derivations. The resource convention is part of every claim. In particular, no finite-ground algorithm is asserted to decide arbitrary universal presentations.

The planned Paper III develops computational representations and checkable certificates for the project. This paper supplies its general finite-ground reference results. The attached visibility prototype is separate from the public QKF-Certifier implementation; the latter is not assumed to implement every invariant or universal theory considered here.

2 The congruence visibility filtration

Let Σ be a finite finitary signature with at least one constant, and let T be its closed term algebra. Constants have depth zero and

$$\text{dep } f(t_1, \dots, t_k) = 1 + \max_i \text{dep } t_i \quad (k > 0).$$

Write $\equiv_E$ for full provable equality and $\mathbb{I}_E = T/\equiv_E$ for the initial algebra. Unless stated otherwise the language is one-sorted. The finite-ground constructions apply sort by sort to a

finite many-sorted signature; the action example uses the two sorts of Paper I.

Definition 2.1 (Bounded term layers and congruence proofs). Put $T^{(D)} = \{t : \text{dep } t \leq D\}$. For ground E , let $\equiv_D$ be the least equivalence relation on $T^{(D)}$ containing every input equation whose two sides have depth at most D and satisfying

$$s_i \equiv_D t_i \ (1 \leq i \leq k) \implies f(s_1, \dots, s_k) \equiv_D f(t_1, \dots, t_k)$$

whenever both displayed result terms belong to $T^{(D)}$. Thus $\equiv_D$ is congruence on the induced partial term algebra.

Equivalently, all terms in an equality proof DAG must have depth at most D , including its premises and conclusions. Congruence can reuse a proved argument equality without placing every intermediate term of that proof inside the new context. For a finite universal presentation, replace the input equations by all their ground instances with sides of depth at most D . Sequential rewrites receive a separate definition in Section 7.1.

Definition 2.2 (Observed, internal, and horizon equality). For $D \geq d$, define the labelled quotient sets

$$\begin{aligned} I_E^{[d]} &= T^{(d)} / (\equiv_E|_{T^{(d)}}), \\ J_E^{(d)} &= T^{(d)} / \equiv_d, \\ K_E^{(d,D)} &= T^{(d)} / (\equiv_D|_{T^{(d)}}). \end{aligned}$$

They are not asserted to be total subalgebras. There are canonical surjections $J_E^{(d)} \twoheadrightarrow K_E^{(d,D)} \twoheadrightarrow I_E^{[d]}$.

Whenever we write $K_E^{(d,D)} \cong I_E^{[d]}$, we mean that this canonical surjection is bijective, or equivalently that the two equivalence relations on the observed terms agree.

Proposition 2.3 (Eventual visibility). *For every fixed d , the canonical map $K_E^{(d,D)} \rightarrow I_E^{[d]}$ is bijective for all sufficiently large D .*

Proof. There are finitely many observed pairs. Every full equality among them has a finite proof containing finitely many terms. Take the maximum of their depths and d . All these proofs are available by that horizon. $\square$

Definition 2.4 (Layer and pairwise visibility). The exact horizon and its overhead are

$$\delta_E(d) = \min\{D \geq d : \equiv_D|_{T^{(d)}} = \equiv_E|_{T^{(d)}}\}, \quad \omega_E(d) = \delta_E(d) - d.$$

For $d(s, t) = \max\{\text{dep } s, \text{dep } t\}$, put

$$\lambda_E(s, t) = \min\{D \geq d(s, t) : s \equiv_D t\},$$

with $\lambda_E(s, t) = \infty$ if $s \not\equiv_E t$. In particular $\lambda_E(t, t) = \text{dep } t$.

The layer invariant is the finite maximum

$$\delta_E(d) = \max \left(d, \max_{\substack{s, t \in T^{(d)} \\ s \equiv_E t}} \lambda_E(s, t) \right).$$

These are invariants of a *presentation with a fixed proof convention*. Adding a derived shallow equation can change them without changing $\mathbb{I}_E$ or its equality relation on the original terms.

2.1 Finite observation sets and coalescence levels

For a finite query set $\mathcal{Q}$, set $d_{\mathcal{Q}} = \max_{q \in \mathcal{Q}} \text{dep } q$, taking $d_{\emptyset} = 0$, and define

$$\delta_E(\mathcal{Q}) = \min\{D \geq d_{\mathcal{Q}} : \equiv_D|_{\mathcal{Q}} = \equiv_E|_{\mathcal{Q}}\}.$$

Thus $\delta_E(\emptyset) = 0$. In general, $\delta_E(d) = \max\{d, \delta_E(T^{(d)})\}$; when $T^{(d)}$ contains a term of depth exactly d , it is simply the query-set special case.

Proposition 2.5 (First-coalescence inequality). *If s, t, u belong to one full equality class, then*

$$\lambda_E(s, u) \leq \max\{\lambda_E(s, t), \lambda_E(t, u)\}.$$

Proof. Both premise equalities hold at the horizon on the right. Transitivity at that horizon gives the conclusion. $\square$

The nested partitions therefore have a merge-tree description. This resembles an ultrametric hierarchy, but $\lambda_E(t, t) = \text{dep } t$ need not be zero, so λ_E is not an ordinary ultrametric. For a final query block B and any $b \in B$, its coalescence level is

$$\max_{q \in B} \lambda_E(b, q) = \max_{p, q \in B} \lambda_E(p, q).$$

The equality follows by the proposition, with endpoint activation included.

Notation	Meaning
$d, D, h(E)$	Observation depth, proof horizon, maximum input-side depth.
λ_E, δ_E	Congruence-proof visibility; the finite-ground algorithm computes these.
$\lambda_E^{\text{rw}}, \delta_E^{\text{rw}}$	Sequential rewrite visibility, defined in Section 7.1.
$\mathcal{Q}, S, S_D$	Query terms, their fixed input/subterm DAG, and its active nodes.
$\approx_D^S$	Congruence closure on the active partial DAG.
$P_D(E, \mathcal{Q})$	The partial quotient $S_D / \approx_D^S$.
$Q_E, \mathcal{I}$	The carrier quotient and named action interface of Paper I.

3 Finite ground presentations

From now through Section 4, let E be a finite set of ground equations.

3.1 Ground locality

Define

$$h(E) = \max\{\text{dep } u : u \text{ is a side of an equation in } E\},$$

with $h(\emptyset) = 0$.

Theorem 3.1 (Ground-locality bound). *For every $d \geq 0$,*

$$\boxed{\delta_E(d) \leq \max\{d, h(E)\}}.$$

Equivalently, if $H = \max\{d, h(E)\}$, then

$$K_E^{(d, H)} \cong I_E^{[d]}.$$

In particular,

$$d \geq h(E) \implies J_E^{(d)} \cong I_E^{[d]}.$$

Proof. Fix $s, t \in T^{(d)}$ with $E \vdash s = t$. Let S be the finite term DAG containing all subterms of the equation sides in E and all subterms of s, t . Ground congruence closure on this induced partial term algebra is complete for the query [3, 4, 5]. Every node in S has depth at most $H = \max\{d, h(E)\}$. Therefore the same equality is generated inside $T^{(H)}$, so $s \equiv_H t$. Soundness gives the converse inclusion. $\square$

Thus congruence visibility in a finite ground presentation has zero overhead at every depth $d \geq h(E)$. This statement concerns proof DAGs; it does not assert the same bound for sequential rewrites.

3.2 Monotonicity

Proposition 3.2 (Monotonicity). *For every presentation for which the visibility depths are defined,*

$$\boxed{\delta_E(d) \leq \delta_E(d+1)}.$$

Proof. Put $D = \delta_E(d+1)$. Equality on $T^{(d+1)}$ is exact at horizon D . Restricting that exact relation to $T^{(d)}$ shows that the smaller layer is exact by the same horizon. $\square$

For a finite ground presentation of exact axiom depth h , define the low-depth profile

$$\mathbf{d}(E) = (\delta_E(0), \dots, \delta_E(h-1))$$

and the overhead fingerprint

$$\Omega(E) = (\omega_E(0), \dots, \omega_E(h-1)).$$

By Theorem 3.1, $\omega_E(d) = 0$ for $d \geq h$.

3.3 The Catalan visibility spectrum

Theorem 3.3 (Catalan visibility spectrum). *Fix $h \geq 1$. The possible low-depth profiles of finite ground equational presentations of exact axiom depth h are exactly the weakly increasing integer sequences*

$$a_0 \leq a_1 \leq \dots \leq a_{h-1}$$

satisfying

$$i \leq a_i \leq h \quad (0 \leq i < h).$$

Consequently there are exactly

$$\boxed{C_{h+1} = \frac{1}{h+2} \binom{2h+2}{h+1}}$$

such profiles.

Proof. Necessity follows from $i \leq \delta_E(i)$, Theorem 3.1, and Proposition 3.2.

For realization, use one unary operation f and finitely many constants. Given an admissible sequence $a = (a_0, \dots, a_{h-1})$, for every i with $a_i > i$ introduce fresh constants p_i, q_i, r_i and the equations

$$f^i(p_i) = f^{a_i}(r_i), \quad f^i(q_i) = f^{a_i}(r_i). \quad (3.1)$$

If none of these equations has depth h , add an isolated equation $f^h(u) = f^h(v)$ on fresh constants to make the axiom depth exactly h .

Fix $d < h$. If $a_d > d$, the terms $f^d(p_d)$ and $f^d(q_d)$ are equal in the full theory through the common witness $f^{a_d}(r_d)$. At any horizon $D < a_d$, neither generating equation of gadget d is admitted, and no other equation mentions its fresh constants. Thus $\delta(d) \geq a_d$. If $a_d = d$, the same lower bound is tautological.

For exactness, the nontrivial full classes within gadget i are

$$\{f^{i+k}(p_i), f^{i+k}(q_i), f^{a_i+k}(r_i)\} \quad (k \geq 0).$$

All remaining terms on its three unary rays are singletons. Fresh constants keep different gadgets disjoint. A gadget with $i > d$ therefore has no nontrivial effect on the observed layer. For $i \leq d$, first derive $f^i(p_i) = f^i(q_i)$ using the witness at depth $a_i \leq a_d$. Congruence then gives their descendants at depth at most $d \leq a_d$, without lifting the witness through every context. If an observed pair also uses the r_i -ray, that witness descendant itself has depth at most d , and the corresponding contextual input equation stays within a_d . The optional isolated depth- h equation has no effect below h . This proves $\delta_E(d) \leq a_d$.

Finally, put $\nu_{i+1} = h - a_i$. The admissibility conditions identify $(\nu_1, \dots, \nu_h)$ with a partition contained in the staircase $(h, h-1, \dots, 1)$. Such partitions are in bijection with Dyck paths of semilength $h+1$, and are counted by C_{h+1} . $\square$

For $h = 2$ there are exactly five profiles:

$$(\delta(0), \delta(1)) \in \{(0, 1), (0, 2), (1, 1), (1, 2), (2, 2)\}.$$

Equivalently,

$$\Omega(E) \in \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 1)\}.$$

3.4 Eliminating finite-ground overhead by derived equations

Proposition 3.4 (Finite visibility flattening). *Every finite ground presentation E has a finite ground extension E^+ with the same full equality relation and $\delta_{E^+}(d) = d$ for every d .*

Proof. Let $h = h(E)$ and adjoin every true equality between terms in $T^{(h)}$. There are finitely many, and they are effectively obtainable by ground congruence closure. Each true pair of depth at most $d \leq h$ is now an input equation active by d . For $d > h$, ground locality already gives exactness at d . The added equations are consequences of E , so the full equality relation is unchanged. $\square$

This existence result does not make the extension small. The relevant optimization question concerns how many derived equations suffice, or how much represented syntax they require, rather than whether a zero-overhead finite ground presentation exists at all.

3.5 Pairwise bottlenecks

Give a reflexivity proof of $t = t$ cost $\text{dep } t$. Assign to an input equation $u = v$ the cost

$$\max\{\text{dep } u, \text{dep } v\}.$$

Combine transitive proof steps by maximum, congruence steps by the maximum of the argument costs and endpoint depths, and alternative derivations by minimum. Let $\mu_E(s, t)$ be the resulting minimum bottleneck cost.

Proposition 3.5 (Bottleneck characterization). *If $E \vdash s = t$, then*

$$\boxed{\mu_E(s, t) = \lambda_E(s, t)}.$$

Proof. For each integer threshold D , the proof objects of bottleneck cost at most D generate exactly the congruence closure of the input equations and subterms admitted at horizon D . Therefore $\mu_E(s, t) \leq D$ if and only if $s \equiv_D t$. Taking the least such threshold gives the claim. $\square$

This does not introduce a new congruence-closure decision procedure. Its point is that the presentation invariant λ_E is exactly a bottleneck optimization problem on standard equality proofs.

4 Exact computation and certificates

4.1 The query-subterm theorem

Let $\mathcal{Q}$ be a finite set of ground terms. Put

$$S = \text{Sub}(E) \cup \text{Sub}(\mathcal{Q}), \quad S_D = \{u \in S : \text{dep } u \leq D\},$$

and let E_D contain exactly the input equations whose sides both have depth at most D . Denote by $\approx_D^S$ the least congruence of the partial term algebra on S_D containing E_D . Here an operation tuple is represented only when the corresponding parent term belongs to S_D .

Theorem 4.1 (Query-subterm visibility). *For $s, t \in \mathcal{Q}$ with $d(s, t) \leq D$,*

$$\boxed{s \equiv_D t \iff s \approx_D^S t.}$$

If $H = \max\{h(E), d_{\mathcal{Q}}\}$, then $\approx_H^S \upharpoonright_{\mathcal{Q}} = \equiv_E \upharpoonright_{\mathcal{Q}}$.

Proof. Every merge on S_D is an admissible bounded equational inference, so the right-hand side implies the left-hand side. For the converse, take the classes of $\approx_D^S$ as a domain. Interpret every represented operation tuple by its parent class; congruence makes this well defined. Complete all unrepresented tuples and constants by a fixed default element. If there are no classes, use a singleton domain instead. Every active term evaluates to its own class, by structural induction, and this algebra satisfies E_D . Consequently two different classes cannot be equal in any equational proof from E_D , including a proof in $T^{(D)}$. This proves the converse. At H , all input equations and queries are present; ground locality gives the final assertion. $\square$

This proof also explains why filling the entire bounded term universe is unnecessary. The finite partial algebra

$$P_D(E, \mathcal{Q}) = S_D / \approx_D^S$$

is determined up to its canonical term labels by the input and query DAG. It is a partial algebra on represented tuples, not a claim that the query set is closed under the signature. The same construction works sort by sort, adding one default element to every empty sort.

4.2 Activation and first merges

Give each node activation level $\text{act}(u) = \text{dep } u$ and each equation activation level $\text{act}(u = v) = d(u, v)$. Activate levels in increasing order, saturating congruence closure at each level before proceeding. The resulting partitions form a nested family on the growing active sets.

A union-find implementation uses parent lists and a signature table keyed by $(f, [u_1], \dots, [u_k])$. Only affected parent signatures need reconsideration after a merge. Classical congruence-closure algorithms provide the relevant efficiency bounds, with the precise bound depending on representation, arity, and proof bookkeeping [4, 7]. We claim no new

asymptotic bound for congruence closure itself. The reduction here is to input subterms with activation labels.

Corollary 4.2 (First-merge characterization). *For final-equal query terms,*

$$\lambda_E(s, t) = \min\{D \geq d(s, t) : s \approx_D^S t\}.$$

For nonempty $\mathcal{Q}$, $\delta_E(\mathcal{Q})$ is the first threshold $D \geq d_{\mathcal{Q}}$ whose query partition equals its final partition at H . For $\mathcal{Q} = \emptyset$, it is zero.

Successful merges can be stored as a levelled forest instead of retaining all partitions. Proposition 2.5 describes the first-coalescence levels in this forest, with endpoint activation accounting for the nonzero diagonal.

4.3 Chronological proof DAGs

When a congruence merge identifies $f(s_1, \dots, s_k)$ with $f(t_1, \dots, t_k)$, its argument equalities already hold. Freeze paths proving those equalities in the current union forest *before* adding the parent merge. Every premise event then has a smaller event number. Reconstructing premises from the final graph alone would lose this guarantee.

Proposition 4.3 (Acyclic depth-optimal certificates). *An activation-ordered, completely saturated run with chronological premise capture yields, for every equal query pair s, t , an acyclic proof DAG whose maximum term depth, including the endpoints, is exactly $\lambda_E(s, t)$.*

Proof. Input equations give leaves; congruence premises refer only to earlier successful merges. Recursive expansion therefore terminates and supplies valid equational inferences. If the first merge of s, t occurs at D , the proof uses only terms active by D . A proof using smaller maximum depth would imply a merge at an earlier completed threshold, contradicting Corollary 4.2. For a reflexive query, the endpoint itself gives the required cost $\text{dep } s$. $\square$

Chronology proves acyclicity. Optimality additionally uses complete saturation and the first-merge theorem; chronology alone is not a lower-bound certificate. Proof-producing congruence closure is established work [6, 7]. Our objective is maximum represented term depth, distinct from explanation cardinality and proof size [8, 9].

4.4 Finite lower-horizon models

Theorem 4.4 (Threshold model). *For every threshold D there is a finite total algebra $M_D \models E_D$ in which each active DAG term evaluates to its $\approx_D^S$ -class. In particular, distinct active classes have distinct values. A nonempty sort needs at most one element per active class; an empty sort is supplied with one default element.*

Proof. Use the represented-tuple interpretation and default completion in the proof of Theorem 4.1. The induction there proves the evaluation assertion and satisfaction of every equation of E_D . $\square$

Corollary 4.5 (Two-sided optimality certificate). *If $\lambda_E(s, t) = D > d(s, t)$, a depth- D proof and a finite model of E_{D-1} separating s, t certify the exact value D . If $D = d(s, t)$, endpoint depth itself gives the lower bound.*

A checker of the positive certificate verifies the actual input equations, terms, inference rules, and premise order. A checker of the lower certificate must evaluate *all* equations of E_{D-1} , not merely those used in the positive proof. The model certifies a bounded lower bound: it need not satisfy deeper input equations. At the final horizon H , however, all of E is active.

Corollary 4.6 (Effective finite separation). *The initial algebra of a finite ground presentation is residually finite: any unequal ground terms are separated by an effectively constructible finite model of that presentation.*

Proof. Apply Theorem 4.4 at $H = \max\{h(E), d(s, t)\}$ to the DAG of E, s, t . The resulting evaluation homomorphism from the initial algebra separates the two term classes. $\square$

4.5 Subset-minimal supports

Collect the input equations used by a depth-optimal proof, and repeatedly test whether one can be deleted while preserving the target equality at that depth. Stop when no deletion succeeds. The remaining set is subset-minimal. Its optimal depth remains the full-input optimum: a smaller-depth proof from a subset would also be a smaller-depth proof from the full input. This procedure does not promise minimum cardinality; small conflict-set optimization is a separate problem [8].

5 The action interface of Paper I

5.1 One example across the series

Paper I studies a finite carrier algebra A , named operators B , a fresh action symbol α , and ground compatibility with the native operations of A . Supplied observations prescribe only selected action values. The initial algebra forces a carrier congruence θ_E^A , a quotient $Q_E = A/\theta_E^A$, and possibly external row values. We now ask at which proof horizon these consequences become visible.

Use the example of Paper I, Example 6.2 [1]. The carrier is $A = \{0, 1, 2, 3\}$, with one binary operation given by

f	0	1	2	3
0	0	2	0	0
1	1	2	1	3
2	0	2	2	0
3	3	0	3	0

and one named operator b . Write $r_i = \alpha(b, i)$ and prescribe

$$r_0 = 0, \quad r_2 = 2, \quad r_3 = 0.$$

The input E consists of the sixteen table equations, these three observations, and sixteen compatibility equations

$$\alpha(b, f(i, j)) = f(r_i, r_j) \quad (i, j \in A).$$

It is a two-sorted presentation, with $f : A^2 \rightarrow A$, b of operator sort, and $\alpha : B \times A \rightarrow A$. Its axiom depth is two. Observe the named interface

$$\mathcal{I} = \{0, 1, 2, 3, r_0, r_1, r_2, r_3\}.$$

At horizon one, the table and observations give exactly five interface classes:

$$\{0, r_0, r_3\}, \quad \{1\}, \quad \{2, r_2\}, \quad \{3\}, \quad \{r_1\}.$$

At horizon two, compatibility gives the proofs

$$\begin{aligned} r_2 &= \alpha(b, f(0, 1)) = f(r_0, r_1) = f(r_3, r_1) = \alpha(b, f(3, 1)) = r_0, \\ r_1 &= \alpha(b, f(1, 0)) = f(r_1, r_0) = f(r_1, r_3) = \alpha(b, f(1, 3)) = r_3. \end{aligned}$$

Every term displayed has depth at most two. Thus $0 = 2$ and $r_1 = 0$ become visible at horizon two. The exact final interface classes are

$$\{0, 2, r_0, r_1, r_2, r_3\}, \quad \{1\}, \quad \{3\}.$$

To exclude further identifications, take the three-element quotient with carrier classes $\{0, 2\}, \{1\}, \{3\}$ and the induced native operation. The constant action row with value $[0]$ is an endomorphism because $f([0], [0]) = [0]$. It satisfies all observations and compatibility equations and separates the three displayed classes.

A lower-horizon model is equally explicit. Take carrier domain $\{0, 1, 2, 3, e\}$, retain the native table on A^2 , define every other f -tuple to have value 0, and set

$$\alpha(b, 0) = 0, \quad \alpha(b, 1) = e, \quad \alpha(b, 2) = 2, \quad \alpha(b, 3) = 0, \quad \alpha(b, e) = 0.$$

With singleton operator domain this is a total model of E_1 . It separates 0 from 2 and r_1 from 0, proving that both equalities require horizon two. In the notation of this paper,

$$\boxed{\lambda_E(0, 2) = \lambda_E(r_1, 0) = \delta_E(\mathcal{I}) = 2.}$$

The source package checks both the symbolic proofs and these finite models.

5.2 Interface size, horizon, and feedback rounds

Let N_D count interface classes at horizon D . In the example $N_1 = 5$ and $N_2 = 3$. More generally, the named-interface formulas of Paper I, Theorem 6.6, compute the two partitions from carrier quotients and row kernels. Since every interface term has depth at most one and every raw compatibility equation has depth at most two, ground locality implies

$$\delta_E(\mathcal{I}) = \begin{cases} 1, & N_1 = N_2, \\ 2, & N_1 > N_2, \end{cases}$$

provided the interface contains an action term. Equality of these finite class counts suffices because the partition at horizon two is a coarsening of that at horizon one.

This depth parameter is separate from the number of quotient-feedback rounds used to compute the fixed point θ_E^A . Paper I, Proposition 5.5, gives a sharp $|A| - 1$ bound on strict feedback rounds, while the proof horizon of this ground signature remains two. Many successive consequences may be proved using terms of the same depth.

5.3 The five depth-two profiles

A second finite example realizes every profile from Theorem 3.3 at $h = 2$. Use a *one-sorted* signature with four distinct constants c_0, c_1, b_0, b_1 , native symbols $+$, $-$, $\cdot$, and a fresh binary symbol α . The native equations are exactly the named tables of addition and negation on $\{c_0, c_1\} \cong \mathbb{Z}/2\mathbb{Z}$, and multiplication on $\{b_0, b_1\} \cong \{0, 1\}$. There is no equation identifying a b -name with a c -name.

Include ground compatibility of each named operator with $+$ and $-$ on named carrier inputs. Set

$$\begin{aligned} \text{Ann} : \quad & \alpha(b_0, c_i) = c_0 \quad (i = 0, 1), \\ R : \quad & \alpha(b_0, c_0) = c_1, \\ \text{Comp} : \quad & \alpha(b_g \cdot b_h, c_i) = \alpha(b_g, \alpha(b_h, c_i)) \quad (g, h, i \in \{0, 1\}). \end{aligned}$$

No additional action unit law is imposed.

Proposition 5.1 (Visibility square). *For the full one-sorted observed layers, these inputs give*

<i>Additional equations</i>	$(\delta(0), \delta(1))$	$ K^{(1,1)} $	$ K^{(1,2)} $
Ann	(0, 1)	44	44
Ann + Comp	(0, 2)	44	43
Ann+ R	(1, 1)	26	26
Ann+ R +Comp	(1, 2)	26	25
<i>Complete flip table</i>	(2, 2)	42	25

In the last row replace Ann by $\alpha(b_g, c_i) = c_{1-i}$ for every g, i , and include neither R nor Comp.

Proof. The equation R creates a horizon-one collision $c_0 = c_1$. Composition adds the delayed equality $\alpha(b_1, c_0) = c_0$ via the instance $(g, h, i) = (1, 0, 0)$. For the flip table, compatibility with $c_1 + c_1 = c_0$ forces $c_0 = c_1$ at depth two. Exactness and absence of further lower-level changes can be checked on a finite DAG by Theorem 4.1: there are four depth-zero terms and $4 + 4 + 3 \cdot 4^2 = 56$ depth-at-most-one terms. Congruence closure on all input/query subterms gives the class counts above and the indicated depth-zero partitions. The accompanying verifier reproduces this finite calculation and independently compares it with closure on the full bounded term universes through depth two. Ground locality makes the final horizon conclusive. $\square$

The square separates a shallow value conflict from a delayed structural consequence. The shared four-element example, in contrast, connects those proof horizons to the quotient and external-row geometry developed in Paper I.

6 Reproducible finite validation

The source supplement contains a Python prototype, a reference bounded-universe closure engine, and deterministic checks for the examples in this paper. The prototype implements congruence visibility on finite ground input. It is a research implementation of the results in Section 4; it is not an implementation of arbitrary universal word-problem search or of semigroup space.

The article-specific verifier reconstructs each input from its displayed definition. It checks both strict resource gaps, all 195 admissible Catalan profiles for $1 \leq h \leq 5$, fourteen visibility-flattening extensions, the shared action example, and the five-case visibility square. For the Catalan constructions it compares 1,077 threshold partitions with an independent full bounded-universe closure. The square contributes fifteen further partition comparisons, using bounded universes of sizes 4, 56, and 9,468. Its query-subterm DAGs contain only 80 or 96 nodes. These are instance sizes, not a claim of a measured runtime speedup.

Validation family	Result of the packaged run
Catalan realization	2, 5, 14, 42, 132 profiles for $h = 1, 2, 3, 4, 5$.
Common action example	$N_1 = 5$, $N_2 = 3$; both target depths equal two.
Strict proof-format gaps	Congruence/rewrite depths 2/3 and 4/5.
Fast versus reference DAG closure	1,000 deterministic random presentations.
Ground DAG versus bounded universe	500 deterministic random presentations.
Chronological proof validation	300 presentations; 3,418 equal queries.
Finite-model construction	200 deterministic random presentations.
Two-sided certificates	81 delayed equalities; 3,662 final nonequalities.
Merge-forest stabilization	500 deterministic random presentations.

All listed checks passed in the packaged run under Python 3.12.14. The supplement records exact commands, random seeds in the scripts, JSON results, finite model tables, and proof explanations. The explicit typed models of Section 5 are evaluated separately from congruence closure. The prototype validates explanations generated from its own closure run; it does not define a standalone interchange format for certificates supplied by an external program. The generic prototype represents one-sorted ground syntax; the mathematical many-sorted extension follows the sortwise construction in Theorem 4.4.

These checks are reproducible evidence for finite instances and implementation invariants. The general spectrum, word-problem, and contextual-envelope theorems are supported by the proofs in the text; they have not been formalized in a proof assistant.

7 Sequential rewrites and universal presentations

7.1 Two bounded proof resources

For a finite set $\mathcal{E}$ of universally quantified identities, an elementary rewrite has the form

$$C[\ell\sigma] \longleftrightarrow C[r\sigma], \quad (\ell = r) \in \mathcal{E},$$

where σ is a ground substitution and C a one-hole ground context. Ground equations are the special case without variables. Define

$$\lambda_{\mathcal{E}}^{\text{rw}}(s, t) = \min_{s=t_0 \leftrightarrow \dots \leftrightarrow t_k = t} \max_{0 \leq i \leq k} \text{dep } t_i,$$

with value ∞ for unequal terms and value $\text{dep } s$ for the empty chain from s to itself. Define $\delta_{\mathcal{E}}^{\text{rw}}(d)$ as the least $D \geq d$ at which such chains recover full equality on $T^{(d)}$. This definition bounds every *whole intermediate term*.

Unrestricted equational proofs can be sequentialized into contextual rewrites [2]. At a fixed horizon this conversion need not preserve maximum depth. Every bounded rewrite is a bounded congruence inference, hence

$$\lambda_{\mathcal{E}}(s, t) \leq \lambda_{\mathcal{E}}^{\text{rw}}(s, t), \quad \delta_{\mathcal{E}}(d) \leq \delta_{\mathcal{E}}^{\text{rw}}(d). \quad (7.1)$$

Both resources converge to the same full equality relation, but the inequalities can be strict.

Example 7.1 (A strict finite-ground gap). Let $E = \{p = f^2(r), q = f^2(r)\}$, with three distinct constants and one unary symbol. Then

$$\lambda_E(f(p), f(q)) = 2, \quad \lambda_E^{\text{rw}}(f(p), f(q)) = 3.$$

At horizon two, derive $p = q$ through $f^2(r)$ and apply congruence to the two depth-one endpoints. No nontrivial input equation is active below two. A sequential rewrite out of $f(p)$, however, must produce $f^3(r)$; the chain $f(p) \leftrightarrow f^3(r) \leftrightarrow f(q)$ attains depth three. Thus even the ground-locality bound of Theorem 3.1 does not transfer to δ^{rw} .

This distinction also shows how presentation-sensitive the resources are. Adding the derived equation $p = q$ preserves the full theory but lowers its pairwise congruence visibility from two to zero. Proposition 3.4 gives a general finite-ground version of this effect.

7.2 Computability and the word problem

For congruence visibility of a fixed finite universal presentation $\mathcal{E}$, use all ground instances whose two sides have depth at most D , and close them in the partial algebra $T^{(D)}$. These

instances are effectively finite: every variable occurring in an identity must be instantiated by a term of depth at most D . Variables not occurring in the identity need not be enumerated. Bounded rewrite equality is likewise computable by reachability in the finite graph of terms of depth at most D .

Both visibility functions are finite at every d , by taking the maximum over finitely many full-equal pairs in $T^{(d)}$ and choosing a finite proof or rewrite chain for each pair. Write $\text{WP}(\mathcal{E})$ for the ground word problem.

Theorem 7.2 (Visibility–word-problem equivalence). *For a fixed finite presentation over a finite signature, the following are equivalent:*

- (1) $\text{WP}(\mathcal{E})$ is decidable;
- (2) $\delta_{\mathcal{E}}$ is computable;
- (3) $\delta_{\mathcal{E}}$ has a computable majorant;
- (4) $\delta_{\mathcal{E}}^{\text{rw}}$ is computable;
- (5) $\delta_{\mathcal{E}}^{\text{rw}}$ has a computable majorant.

Proof. For either resource, a computable majorant gives a decision algorithm: on s, t , compute the corresponding bounded relation at that majorant of $d(s, t)$. Conversely, a ground word-problem algorithm computes the exact equality partition of the finite layer $T^{(d)}$. Enumerate horizons $D = d, d + 1, \dots$ and compute the bounded partitions until one equals that full partition. Eventual exactness guarantees termination, and the first matching horizon is the relevant visibility value. Computability implies the existence of a computable majorant in each case. $\square$

Thus either function is a modulus of convergence to the ground word problem. For each fixed finite universal presentation with undecidable ground word problem, neither function has a computable majorant. The finite-ground bound depends essentially on the equations being ground; syntactic depth of universal axioms alone gives no comparable bound.

8 Semigroup derivational space inside action chains

8.1 Clean action chains

Let

$$P = \langle G \mid R \rangle$$

be a finite semigroup presentation with nonempty finite generator set G and defining relations between nonempty words. Use a one-sorted signature consisting of a base constant a , one constant b_g for each generator $g \in G$, and a single binary operation α . Set $W_{\varepsilon}(x) = x$. For a nonempty word $w = g_1 \cdots g_k$, define

$$W_w(x) = \alpha(b_{g_1}, \alpha(b_{g_2}, \dots, \alpha(b_{g_k}, x) \dots)). \quad (8.1)$$

Let $\mathcal{A}(P)$ consist of the universal identities

$$W_u(x) = W_v(x) \quad ((u, v) \in R). \quad (8.2)$$

Theorem 8.1 (Clean-chain correspondence). *For nonempty words u, v ,*

$$\boxed{u =_P v \iff \mathcal{A}(P) \vdash W_u(a) = W_v(a).}$$

Moreover, elementary applications of a defining semigroup relation correspond exactly to elementary equational rewrites between clean chains.

Proof. A factor replacement $puq \leftrightarrow pvq$ is obtained by substituting $x \mapsto W_q(a)$ into the identity $W_u(x) = W_v(x)$ and placing the result in the outer context $W_p[-]$.

Conversely, every α -rooted subterm of a clean chain is a suffix chain. Any instance of a defining identity that matches a clean chain therefore substitutes a clean suffix for x , and the surrounding context is a clean prefix. Hence every elementary equational rewrite projects to one semigroup factor replacement. Unrestricted equational equality is generated by such rewrites, proving the reverse implication as well. $\square$

For equal words define

$$\begin{aligned} \text{sp}_P(u, v) &= \min_{u=w_0 \leftrightarrow \dots \leftrightarrow w_k=v} \max_i |w_i|, \\ s_P(n) &= \max_{\substack{u \equiv_P v \\ |u|, |v| \leq n}} \text{sp}_P(u, v). \end{aligned}$$

Trivial pairs are included, so $s_P(n) \geq n$ for $n \geq 1$. The maxima exist because there are only finitely many endpoint words of bounded length. This is the derivational-space function of the finite semigroup presentation [12]. On clean chains, syntactic depth is word length, so Theorem 8.1 yields

$$\boxed{\lambda_{\mathcal{A}(P)}^{\text{rw}}(W_u(a), W_v(a)) = \text{sp}_P(u, v).} \quad (8.3)$$

Example 8.2 (Congruence reuse does not measure word space). Take $P = \langle p, x, y, z \mid x = zzzz, y = zzzz \rangle$. Then

$$\lambda_{\mathcal{A}(P)}(W_{px}(a), W_{py}(a)) = 4, \quad \text{sp}_P(px, py) = \lambda_{\mathcal{A}(P)}^{\text{rw}}(W_{px}(a), W_{py}(a)) = 5.$$

For the first equality, derive $W_x(a) = W_y(a)$ through the depth-four witness $W_{zzzz}(a)$ and apply congruence with first argument b_p . Below depth four no nontrivial ground instance of a defining identity is available. For the second equality, the first rewrite from px must produce $pzzzz$, of length five, and $px \leftrightarrow pzzzz \leftrightarrow py$ attains that bound. Thus the exact clean-chain correspondence with space cannot use λ in place of λ^{rw} .

8.2 The typed variant

In the natural two-sorted variant

$$\alpha : \text{Op} \times \text{State} \rightarrow \text{State},$$

with generator constants of sort **Op** and one state constant, every closed state term is already a clean chain. There is no skeleton padding and

$$\delta^{\text{rw}}(n) = s_P(n)$$

exactly for $n \geq 1$, while $\delta^{\text{rw}}(0) = 0$. Generator-sort terms are immutable constants; the isolated base state contributes only a reflexive query. This pointwise identity concerns sequential rewrites.

8.3 Skeleton decomposition of arbitrary one-sorted terms

Call a node of a ground term *marked* when it is literally of the form $\alpha(b_g, q)$ for some generator g . A *chain block* is a maximal consecutive sequence of marked nodes obtained by repeatedly following the second child. Each block carries a nonempty generator word. Collapse every maximal word to an abstract block label while preserving its location, tail, every unmarked α -node, and all constants. The resulting tree is the *skeleton*.

Lemma 8.3 (Skeleton invariance). *One elementary rewrite using a defining identity in $\mathcal{A}(P)$ preserves the skeleton, preserves all chain-block words except one, and changes that one word by exactly one semigroup factor replacement. Conversely, every defining factor replacement inside a chain block is realized by one equational rewrite.*

Proof. A match of $W_u(x)$ can start only at a marked node and follows marked second-child nodes through the nonempty word u . The substitution for x is the suffix following the matched factor. Replacing u by v therefore changes only that factor of the same maximal block.

No rewrite changes an unmarked node into a marked one or conversely. Constants are immutable. A nonconstant first child remains nonconstant because both sides of every defining identity are α -rooted, and a non-generator constant remains itself. Hence the marked/unmarked status of every surrounding node is invariant. The converse is the same matching observation read backwards. $\square$

Corollary 8.4 (Global equality decomposition). *Two arbitrary ground action terms are equal modulo $\mathcal{A}(P)$ if and only if they have the same skeleton and the words on every pair of corresponding chain blocks are equal in P .*

Thus the one-sorted ground word problem is a syntactic skeleton comparison followed by finitely many semigroup word-problem instances.

8.4 The contextual-envelope theorem

Theorem 8.5 (Global rewrite-visibility envelope). *For every $n \geq 1$,*

$$\delta_{\mathcal{A}(P)}^{\text{rw}}(n) = \max_{1 \leq m \leq n} (n - m + s_P(m)) \quad (8.4)$$

and hence

$$\delta_{\mathcal{A}(P)}^{\text{rw}}(n) - n = \max_{1 \leq m \leq n} (s_P(m) - m). \quad (8.5)$$

Proof. For the lower bound, fix $m \leq n$ and equal words u, v realizing $s_P(m)$. The base constant a is syntactically distinct from all generator constants. Define frozen contexts

$$C_0[x] = x, \quad C_{k+1}[x] = \alpha(a, C_k[x]).$$

Every outer node is unmarked, so no defining identity matches at its root. Any sequential rewrite chain between

$$C_{n-m}[W_u(a)] \quad \text{and} \quad C_{n-m}[W_v(a)]$$

therefore performs a semigroup derivation inside a frozen context. Its maximum term depth is $n - m$ plus the maximum intermediate word length. Hence the pair has rewrite visibility depth $n - m + s_P(m)$. Maximizing over m gives the lower bound.

For the upper bound, let equal ground terms s, t have depth at most n . By Corollary 8.4, they have the same skeleton. If there are no chain blocks, the terms are identical and depth n suffices. Otherwise index the corresponding chain blocks by i , with source and target words u_i, v_i , and put

$$m_i = \max\{|u_i|, |v_i|\}.$$

For each block choose a semigroup derivation whose intermediate lengths are at most $s_P(m_i)$.

For a fixed skeleton, total term depth is a monotone function of the block lengths and is 1-Lipschitz in each block length separately: increasing one block by r can increase a root-to-leaf path by at most r .

First process all blocks with $|u_i| \geq |v_i|$. Before and after each conversion the current block-length vector is coordinatewise no larger than the source vector, so the term depth is at most n ; while block i is being rewritten, its temporary increase beyond the baseline length $m_i = |u_i|$ is at most $s_P(m_i) - m_i$. The total depth is therefore at most

$$n - m_i + s_P(m_i).$$

Then process all blocks with $|u_i| < |v_i|$. At this stage the current vector, with unprocessed increasing blocks still at source length, is coordinatewise bounded by the target vector. Comparing the active block with the baseline in which it already has target length $m_i = |v_i|$ gives the same bound. Hence a derivation exists with maximum depth at most

$$\max_i (n - m_i + s_P(m_i)) \leq \max_{1 \leq m \leq n} (n - m + s_P(m)).$$

□

Define the contextual envelope of any function $s(n) \geq n$ by

$$H(s)(n) = n + \max_{1 \leq m \leq n} (s(m) - m). \quad (8.6)$$

Proposition 8.6 (Minimal context-stable envelope). *H is the least majorant of s whose overhead $H(s)(n) - n$ is nondecreasing. It is idempotent:*

$$H(H(s)) = H(s).$$

If $s_P(n) - n$ is already nondecreasing, then

$$\delta_{\mathcal{A}(P)}^{\text{rw}}(n) = s_P(n)$$

pointwise.

Proof. The running maximum in (8.6) is nondecreasing and dominates $s(n) - n$, so $H(s) \geq s$. If $f \geq s$ and $f(n) - n$ is nondecreasing, then for every $m \leq n$,

$$f(n) - n \geq f(m) - m \geq s(m) - m,$$

so $f(n) \geq H(s)(n)$. Minimality gives idempotence. □

Since s_P is nondecreasing and $s_P(n) \geq n$,

$$s_P(n) \leq \delta_{\mathcal{A}(P)}^{\text{rw}}(n) \leq n + s_P(n) \leq 2s_P(n). \quad (8.7)$$

Following the standard equivalence for space functions [12], write $f \preceq g$ if some positive integer c satisfies

$$f(n) \leq c g(cn) + cn$$

for all n , and write $f \simeq g$ when both directions hold. Then

$$\boxed{\delta_{\mathcal{A}(P)}^{\text{rw}} \simeq s_P.} \quad (8.8)$$

8.5 Complexity-spectrum consequences

The exact envelope formula transfers realization results for semigroup space functions.

Corollary 8.7 (Deterministic-space realization). *Let $f(n)$ be the space-complexity function of a deterministic Turing machine. There exists a finite one-sorted equational presentation using only constants, one binary operation α , and finitely many universal action-chain identities such that*

$$\delta_E^{\text{rw}} \simeq f.$$

Proof. Olshanskii proves that such f is equivalent to the space function of a finitely presented semigroup [12, Corollary 1.4]. Apply Theorem 8.5 and (8.8). $\square$

Corollary 8.8. *There exists a finite action-chain presentation with PSPACE-complete ground word problem and polynomial rewrite-visibility function.*

Proof. Olshanskii gives a finitely presented semigroup with PSPACE-complete word problem and polynomial space function [12, Corollary 1.5]. Clean chains give the hardness reduction. Corollary 8.4 reduces arbitrary one-sorted action-term equality to a skeleton comparison and finitely many semigroup word-problem instances, solvable sequentially in polynomial space. Theorem 8.5 gives polynomial rewrite visibility, and (7.1) also bounds congruence visibility from above. $\square$

Olshanskii also determines, up to the same equivalence, which power functions occur as semigroup space functions [12, Corollary 1.6]; the same realization criterion transfers to rewrite-visibility functions of finite action-chain presentations.

These realization statements concern δ^{rw} . They neither identify congruence visibility δ with semigroup space nor identify a presentation resource with the optimal machine-space complexity of its word problem.

8.6 Immediate exactness under depth-preserving identities

At the opposite extreme, call an identity $\ell = r$ *depth-preserving* if every ground substitution σ satisfies

$$\text{dep}(\ell\sigma) = \text{dep}(r\sigma).$$

If every identity of a finite presentation is depth-preserving, then every elementary rewrite preserves the depth of the whole term in every context. Therefore no derivation starting in $T^{(d)}$ needs to leave it, and

$$\boxed{\delta_E(d) = \delta_E^{\text{rw}}(d) = d \quad \text{for all } d.}$$

This contrasts with action-chain relations of unequal word length, whose rewrite visibility can realize the semigroup-space behavior above.

9 Relation to established methods

The finite-ground construction uses classical congruence closure. Nelson–Oppen and Downey–Sethi–Tarjan established foundational algorithms for ground equality [3, 4]; Kapur related congruence closure to completion [5]. The activation filtration records when the equations and represented terms needed by these procedures become available. The locality theorem is the classical subterm property expressed as a depth bound; the Catalan theorem classifies the possible exact profiles under that bound.

Proof production also has a substantial prior literature [6, 7]. Minimum explanation size and minimum support cardinality differ from maximum term depth [8, 9]. The two-sided certificate here combines an explanation at the first complete threshold with a finite model below it. Local inference and local theory extensions likewise control the instantiated syntax required for ground reasoning, under their respective structural hypotheses [10, 11]. No extension-locality hypothesis is needed for our finite set of ground equations.

For universal presentations, the action-chain construction links sequential rewrite depth with semigroup derivational space. Olshanskii’s invariant counts maximum intermediate word length, and his realization theorems supply the complexity consequences used here [12]. The skeleton argument accounts for arbitrary one-sorted contexts by the explicit overhead envelope H . Example 8.2 marks the limit of that correspondence: congruence proofs can reuse established equalities without carrying their entire intermediate syntax inside the current context.

Paper I supplies the algebraic interpretation of the observed interface and the quotient-feedback mechanism [1]. Here the input terms are arbitrary ground syntax; the connection to incomplete actions is an application. This division lets Paper III cite the query-subterm, first-merge, and threshold-model results directly when describing a concrete certifying implementation.

10 Further questions

- (1) **Compact presentation optimization.** Proposition 3.4 settles existence of a zero-overhead finite ground extension. How many derived equations, or how many distinct subterms, are needed to achieve a given improvement? Which transformations of universal presentations preserve either asymptotic visibility class?
- (2) **Fixed-signature spectra.** The Catalan construction uses fresh constants as independent gadgets. Which profiles survive when the signature, number of constants, or number of equations is fixed in advance?
- (3) **The gap between proof formats.** How large can $\lambda_E^{\text{rw}}(s, t) - \lambda_E(s, t)$ be under restrictions on arity, contexts, or relation shape? For action chains, what invariant describes congruence visibility itself beyond the upper bound by the sequential envelope?
- (4) **Secondary certificate objectives.** Once minimum term depth is fixed, what is the cost of optimizing proof size or support cardinality? How small can a lower-threshold separating model be, compared with the canonical DAG-class construction?
- (5) **Other context geometries.** Ordinary one-sorted action contexts yield $H(s)(n) = n + \max_{1 \leq m \leq n}(s(m) - m)$ for sequential rewrites. Which envelope operators arise under different sort restrictions or other explicitly specified proof resources?

11 Conclusion

Equality visibility records the difference between the depth of a question and the depth required to establish its answer in a specified proof format. For congruence proofs over finite ground input, locality gives a finite exact horizon, all admissible low-depth profiles occur, and their spectrum is Catalan. A single activated input/query DAG computes first-equality levels. Positive proof DAGs and finite lower-threshold models together certify their exactness.

The common action example connects this filtration to Paper I: the named interface has five classes at horizon one and three at horizon two, while quotient-feedback rounds remain a separate computational parameter. These finite-ground results form the reference layer for the planned algorithm paper.

For universal presentations, computable bounds for either proof format are equivalent to decidability of the ground word problem. Sequential action-chain visibility has a further exact description by semigroup space and its contextual envelope. Keeping the sequential and congruence invariants distinct preserves that correspondence and specifies which guarantees the finite-ground algorithm actually computes.

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